ÁKOS SZABÓ-SZIKSZAY

Data Science and Artificial Intelligence Developer

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Portfolio: akosszsz.github.io/aboutme/ | **GitHub:** https://github.com/AkosSzSz



PROFESSIONAL SUMMARY

I am an 11th-grade secondary school student specializing in mathematics and physics, and a data science and AI developer with nearly 6 years of programming experience. The Faculty of Engineering and Information Technology (GAMF) of John von Neumann University awarded me the "Certificate of Merit for the GAMF Faculty" for my AI-based development work on the solar-powered race car that finished 10th in the world at the World Solar Challenge 2025. I independently built a web portfolio and an in-browser algorithm lab that visualizes machine learning and optimization algorithms. I am looking for a junior or intern position where I can contribute to the success of complex AI projects.

AWARDS AND RECOGNITION

"Certificate of Merit for the GAMF Faculty" Award & Project Recognition — Kecskemét, 22 Oct 2025

John von Neumann University, GAMF Faculty of Engineering and Information Technology

- Official university award and certificate for outstanding completion of artificial intelligence and data science-based meteorological development tasks for the World Solar Challenge 2025.
- The award was issued by Dr. Lóránt Kovács, Dean, and László Tóth, Project Leader, for developing the weather-forecast correction software that supported the solar race car's race strategy.

FEATURED PROJECTS & TEACHING EXPERIENCE

Weather Forecast Correction AI and Data Model — 2024 – 2025

Neumann Solar Team – University Development Project

- I designed a data-processing and predictive correction software (a bias correction algorithm) that analyzes the systematic deviations between historical forecasts and real-time measured telemetry data.
- **Algorithmic implementation:** Integration of complex JSON data structures with asynchronous timestamps, delta calculation of hourly meteorological parameters (temperature, humidity, precipitation), and dynamic time-offset correction of sunrise, sunset and the solar angle of incidence for optimal energy use by the car.
- I was responsible for designing and implementing the algorithm; the model supported the team's race strategy, and the Hungarian team finished 10th in the world. (The full production version is protected under university confidentiality.)

Educational Software & Interactive Physics Simulations — 2024 – 2026

Újpest Könyves Kálmán Secondary School – School and Methodology Projects

- **Nuclear Physics Simulation System (Energy Valley project):** I independently designed and developed an interactive educational software and visualization engine that models alpha, beta and gamma decay processes in real time, the convergence of exponential decay curves, and the deflection of particles in a cloud chamber.
- **Programming methodology:** In lessons I taught on my own, I introduced my classmates to Clean Code principles, event-driven programming (Pygame) and persistent data management through a custom-built Snake game.

EXPERIMENTAL LAB (PROJECTS RUNNING DIRECTLY IN THE BROWSER)

I built the entire website and the interactive algorithms in the lab on my own in about 2 weeks, learning from YouTube videos and online literature:

- **K-means clustering:** Centroid selection using the k-means++ method, with step-by-step visual convergence in a JavaScript environment.
- **Neural network:** Visualization of feed-forward computations with continuously changing inputs and a sigmoid activation function.
- **Genetic route optimizer:** A population-based TSP solution built on tournament parent selection, order crossover (OX) and mutation.

TECHNICAL SKILLS

Primary programming language: Python (Advanced, nearly 6 years of experience – OOP, Pygame, data handling)

Other languages: C++, C#, Java (basic level)

Teamwork: Worked on a university development project as a member of the Neumann Solar Team

Web development: JavaScript, HTML (the basis of the portfolio site and the interactive lab)

Tools & Competencies: Git/GitHub, JSON/file handling, predictive modeling, modeling of physical processes

EDUCATION & TRAINING

Újpest Könyves Kálmán Secondary School **2023 – Present**

Advanced-level Mathematics–Physics track (elective course) | Expected graduation (Matura): 2028

Logiscool Budapest - Újpest **2020 – 2024**

Data Science & Artificial Intelligence specialization (2023–2024)

Python professional exam, Python Advanced (432) and Python Master (433) certificates

LANGUAGES

- **English:** Upper-intermediate (B2) language exam certificate | Currently studying at C1+ level at school (learning for 11 years).
- **German:** Pre-intermediate/Intermediate (B1+) level (learning for 3 years) | Goal: B2 language exam next year.